



CHALLENGES & LIMITATIONS

Building a Cross-Platform, Multi-User Motion-Sensing Game for the Living Room

What We're Building

A quick look at the project before we get into the challenges



The Idea

A motion-sensing game where a camera or phone turns your body into the controller -- no extra hardware needed.



The Tech

We track a player's skeleton in real time from video, then feed that motion data into the game.



The Setting

Designed for the living room, where families and friends play together -- not a solo VR headset experience.

Wrong Platform, Wrong Audience



The Problem

We started with Apple's ARKit API, which gives 3D skeletal data built for virtual reality users. But our real goal is a living-room game -- a different audience. ARKit's VR-grade data also only runs on newer, high-end Apple devices.



Our Solution

We switched to cross-platform 2D skeletal data that runs on most standard smartphones. It's less specialized than ARKit, but it lets us reach everyday users in the living room -- not just a small group of tech enthusiasts.

The Training-Data Bottleneck



Our core task is recognizing actions from video. Getting a model good enough for a real product needs a massive amount of training data -- far more than a small team can realistically collect and label from scratch.

Our Solution: Don't Reinvent the Wheel

- We dropped the plan to train a model from zero.
- Instead, we adopted pre-built solutions from major tech companies that already predict skeletal data streams directly from video.
- Result: production-level reliability, without needing a huge in-house dataset.

Challenge 3

One Living Room, Many Players



The Problem

Our first working version was a single-user, single-device game. But living rooms are almost never used by just one person -- families and friends play together. Our system needed to support multiple users at once.



Our Solution

We built a high-performance gateway that lets multiple devices connect at once. Each phone or camera in the room streams its own player's skeletal data into the same game session in real time.

An Honest Limitation We Still Have



One camera or phone can only capture one person's skeleton at a time.

The gateway solves the connection problem -- it can bring many devices into one session -- but it doesn't remove this per-device limit. In practice, this means one device per player is still required: the gateway lets us scale to a full room, but it can't turn a single camera into a multi-person tracker.

Beyond Full-Body Tracking



Where We Are Now

Our model currently captures full-body data only. But a lot of gameplay ideas need more precision than that -- or a completely different kind of tracking altogether.



Where We're Headed

In principle, we can track anything human-related -- a waved game prop, or just finger movement. Our new gateway is built to support different skeletal models, and the games that rely on them.

Putting It All Together

Four challenges, four decisions

Challenge	Our Response
ARKit's 3D data targeted VR users, not our living-room audience	Switched to cross-platform 2D skeletal data for broad phone compatibility
Training an action-recognition model from scratch needs huge datasets	Adopted pre-built skeletal-prediction models from major tech companies
Living rooms have multiple players, but our game was single-user	Built a high-performance gateway supporting multiple connected devices
Full-body-only tracking limits future gameplay ideas	Designed the gateway to support flexible skeletal models (props, fingers, etc.)

Key Takeaways



Design for your real audience

The most advanced technology (ARKit) wasn't the right choice -- the most compatible one was.



Don't reinvent what already exists

Using proven, pre-built models saved us from an unrealistic data-collection effort.



Build flexible infrastructure early

The gateway we built for multi-user support also opened the door to new tracking types.



Thank You

Questions & Discussion